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0 vimrc

vimrc

```

1 "variable
2 let mapleader = ','
3
4 "
5 set number
6 set relativenumber
7 set noexpandtab
8 set shiftwidth=4
9 set tabstop=4
10
11 inoremap kj <Esc>
12 inoremap {<CR> {<Left><CR><Esc>0
13 tnoremap kj <C-]><C-n>
14 noremap <leader>c :w<CR>:!g++ -std=c++20 % -o %:r &&
   ↪ ./%:r<CR>
15 noremap ⌂ ⌂
16 noremap ⌂ ⌂

```

1 Number Theory

Exgcd.cpp

```

1 array<ll,2> exgcd(ll a, ll b){
2     if(b == 0) return array<ll,2>{1,0};
3     auto ans = exgcd(b, a % b);
4     return array<ll,2>{ans[1], ans[0] - a / b *
   ↪ ans[1]};
5 }
6

```

catalan.cpp

```

1 C_0=1
2 C_n = $\sum C_i * C_{n-1-i}$ = $\frac{1}{n+1} C_n^{2n}$

```

findPrime.cpp

```

1 constexpr ll mxN=1e6+1;
2 mobius[1]=etf[1]=1;
3 forn(i,2,mxN){
4     if(!prime[i])
   ↪ prime[i]=i,mobius[i]=-1,etf[i]=i-1,st[++sz]=i;
5     forn(j,1,sz+1){
6         if(i*st[j]>mxN) break;
7         prime[i*st[j]]=st[j];
8         if(i%st[j]==0){
9             mobius[i*st[j]]=0;
10            etf[i*st[j]]=etf[i]*st[j];
11            break;
12        }
13        else mobius[i*st[j]]=mobius[i]*-1,etf[i*st
   ↪ [j]]=etf[i]*(st[j]-1);
14    }
15 }

```

matrixMultiple.cpp

```

1 // mod 需外部定義; 快速幂用時記得單位矩陣
2 vector<vector<ll>> matrixMultiply(const
   ↪ vector<vector<ll>>& A, const vector<vector<ll>>& B) {
3     ll rowA = A.size();
4     ll colA = A[0].size();

```

```

11 colB = B[0].size();
12 vector<vector<ll>> C(rowA, vector<ll>(colB, 0));
13 for (ll i = 0; i < rowA; ++i) {
14     for (ll k = 0; k < colA; ++k) {
15         for (ll j = 0; j < colB; ++j) {
16             C[i][j] = (C[i][j] + A[i][k] * B[k][j]) %
   ↪ mod;
17         }
18     }
19 }
20 return C;
21 }

```

ntt.cpp

```

1 using ll=long long;
2 // 注意: 兩模數CRT, 結果是真值 mod 998244353*1004535809
   ↪ (~1.0028e18)
3 // 真實係數必須 < 該值 (值域1e9且長度>~1000就會爆),
   ↪ 否則需三模數NTT
4 // 長度上限 2^21 (n+m <= ~2e6): fft2 模數只有 2^21 階單位根,
   ↪ 超過會靜默出錯
5 class FFT{
6     public:
7         ll mod,root,root_inv,maX,M,M_inv;
8         FFT(){
9             FFT(ll a,ll b,ll c,ll d,ll e,ll
   ↪ f):mod(a),root(b),root_inv(c),maX(d),M
   ↪ (e),M_inv(f){
10                 ll fsp(ll a,ll m){
11                     ll ans=1;
12                     while(m){
13                         if(m&1) ans=(ans*a)%mod;
14                         a=(a*a)%mod,m>>=1;
15                     }
16                     return ans;
17                 }
18                 void fft(ll sz,vc<ll> &arr,ll inv){
19                     for(ll i=1,j=0;i<sz;++i){
20                         ll bit=sz>>1;
21                         ↪ for(;j&bit;bit>>=1)
22                         ↪ j^=bit; j^=bit;
23                         if(i<j)
24                             ↪ swap(arr[i],arr[j]);
25                     }
26                     for(ll l=2;l<=sz;l<=<=1){
27                         ll wlen=inv?root_inv:root;
28                         for(ll i=l;i<maX;i<=<=1)
29                             ↪ wlen=(wlen*wlen)%mod;
30                         for(ll i=0;i<sz;i+=l){
31                             ll w=1;
32                             for(ll
   ↪ j=0;j<l/2;++j){
33                                 ll x=arr[i
   ↪ +j],y=
   ↪ (w*arr
   ↪ [i+j+l
   ↪ /2])%m
   ↪ od;
34                                 arr[i+j]=
   ↪ (x+y<mo
   ↪ d?x+y:
   ↪ x+y-mo
   ↪ d);
35                                 arr[i+j+l/
   ↪ 2]=(x-
   ↪ y>=0?x
   ↪ -y:x-y
   ↪ +mod);
36                                 w=(w*wlen)
   ↪ %mod;
37                             }
38                         }
39                     if(inv){
40                         ll n_1=fsp(sz,mod-2);

```

```

38         forn(i,0,sz){
39             arr[i]=(arr[i]*n_18)
               ↳ %)mod;
40         }
41     }
42 }
43 };
44 constexpr ll MM=1002772198720536577;
45 ll mul(ll a,ll b){
46     __int128 aa=a,bb=b,cc;
47     cc=(aa*bb)%MM;
48     return (ll)cc;
49 }
50 void solve(){
51     FFT fft1(998244353,15311432,469870224,1<<23,100453589)
               ↳ 5809,332747959);
52     FFT fft2(1004535809,702606812,700146880,1<<21,99830)
               ↳ 44353,669690699);
53     ll n,m;
54     cin >> n >> m;
55     vc<ll> a(n+1),b(m+1);
56     for(auto &v:a) cin >> v;
57     for(auto &v:b) cin >> v;
58     ll sz=1;
59     for(;sz<a.size()+b.size();sz<=<=1);
60     a.resize(sz),b.resize(sz);
61     vc<ll> A=a,B=b;
62     fft1.fft(sz,a,0);
63     fft1.fft(sz,b,0);
64     forn(i,0,sz) a[i]=(a[i]*b[i])%fft1.mod;
65     fft1.fft(sz,a,1);
66
67     fft2.fft(sz,A,0);
68     fft2.fft(sz,B,0);
69     forn(i,0,sz) A[i]=(A[i]*B[i])%fft2.mod;
70     fft2.fft(sz,A,1);
71
72     forn(i,0,n+m+1) cout <<
               ↳ (mul(a[i],fft1.M*fft1.M_inv) +
               ↳ mul(A[i],fft2.M*fft2.M_inv))%MM << "
               ↳ \n"[i==n+m];
73 }
74 int main()
75 {
76     fast_io;
77     int testcase;
78     // cin>>testcase;
79     testcase=1;
80     while(testcase--){
81         solve();
82         return 0;
83     }

```

system-of-linear-eqn.cpp

```

1 // mod mdl1 下高斯消去。無解輸出-1；無窮多解「不會偵測」
   ↳ (自由變數=0給特解)，題目要區分時要自己加
2 // 輸入係數需在 [0,mdl1) 內 (負數要先轉正); inv
   ↳ 其實是快速冪，模數寫死 mdl1
3 constexpr ll mxN=505;
4 ll n,m;
5 ll inv(ll a,ll m){ll ans; for(ans=1;m;ans=(m&1?(ans*a)%mdl1
   ↳ 1:ans),a=(a*a)%mdl1,m>>=1); return
   ↳ ans;}
6 void solve(istream &cin){
7     cin>>n>>m;
8     vc<ll> res(m,0);
9     vc<vc<ll>> a(n,vc<ll>>(m+1));
10    forn(i,0,n) forn(j,0,m+1) cin>>a[i][j];
11    ll l=0;
12    for(ll i=0;i<m && l<n;++i){
13        if(!a[l][i]) forn(j,l+1,n)
               ↳ if(a[j][i]){swap(a[j],a[l]); break;}
14        if(!a[l][i]) continue;
15        ll c=inv(a[l][i],mdl1-2);
16        for(auto&v:a[l]) v=(v*c)%mdl1;

```

```

        forn(j,l+1,n){
            ll co=a[j][i];
            forn(k,i,m+1) a[j][k]=((a[j][k]-co)
               ↳ *a[l][k])%mdl1+mdl1)%mdl1;
        }
        l++;
    }
    forn(i,n-1,0){
        ll j=0;
        while(j<m+1 && !a[i][j]) j++;
        if(j==a[i].size()) continue;
        if(j+1==a[i].size()){cout << -1 << '\n';
               ↳ return;}
        ll u=a[i][m];
        forn(k,j+1,m)
               ↳ u=((u-a[i][k]*res[k])%mdl1+mdl1)%mdl1;
        res[j]=u;
    }
    forn(i,0,m) cout << res[i] << " \n"[i==m-1];
}
int main()
{
    fast_io;
    ll testcase;
    // cin>>testcase;
    testcase=1;
    while(testcase--){
        solve(cin);
        return 0;
    }
}

```

2 Tree

LCA.cpp

```

1 constexpr ll mxN=2e5+1;
2 ll pa[mxN][20],d[mxN]={};
3 vc<ll> adj[mxN];
4 void dfs(ll u=1,ll p=0){d[u]=d[p]+1,pa[u][0]=p;
   ↳ for(auto&v:adj[u]) if(v!=p) dfs(v,u);}
5 void init(ll n){ // 建好 adj 後呼叫一次
6     dfs();
7     forn(j,1,20) forn(i,1,n+1)
               ↳ pa[i][j]=pa[pa[i][j-1]][j-1];
8 }
9 ll lca(ll u,ll v){
10     if(d[u]<d[v]) swap(u,v);
11     forr(i,19,0) if(d[u]-(1<<i)>=d[v]) u=pa[u][i];
12     if(u==v) return u;
13     forr(i,19,0) if(pa[u][i]!=pa[v][i])
               ↳ u=pa[u][i],v=pa[v][i];
14     return pa[u][0];
15 }

```

MST.cpp

```

1 //kruskal
2 //vertex is 1 index
3 #define mls multiset
4
5 ll n,pa[mxN]; //memset(pa,-1,sizeof(pa));
6 mls<ar<ll,3>> adj;
7
8 ll find(ll x){return pa[x]<0?x:pa[x]=find(pa[x]);}
9 void un(ll x,ll y){
10     x=find(x),y=find(y); if(x==y) return;
11     if(-pa[x]>-pa[y]) swap(x,y); pa[y]+=pa[x],pa[x]=y;
12 }
13 ll kruskal(){ // 回傳 MST 總權重；結束後 cnt<n-1 代表圖不連通
14     ll cnt=0,wi=0;
15     while(!adj.empty() && cnt<n-1){
16         auto [w,u,v]=*adj.begin(); adj.erase(adj.begin());

```

```

17         if(find(u)!=find(v)) wi+=w,un(u,v),cnt++;
18     }
19     return wi;
20 }

```

binary-lifing.cpp

```

1  constexpr ll mxN=2e5+1;
2  ll n,q,p[mxN][31]={0};
3  vc<ll> adj[mxN];
4  void solve(istream &cin){
5      cin>>n>>q;
6      forn(i,2,n+1) cin>>p[i][0];
7      forn(j,1,31) forn(i,1,n+1) p[i][j]=p[p[i][j-1]][j-1];
8      while(q--){
9          ll x,k; cin>>x>>k;
10         k=min(k,(ll)mxN); // 跳超過深度必到0,
11         // clamp避免高位bit被忽略
12         forn(i,0,31) if(k>>i&1) x=p[x][i];
13         cout << (x?x:-1) << '\n';
14     }

```

find-centriod.cpp

```

1  void dfs(ll u=1,ll p=0){
2      d[u]=1;
3      for(auto&v:adj[u]){
4          if(v==p)
5              continue;
6          dfs(v,u);
7          d[u]+=d[v];
8      }
9  }
10 ll dfs_solve(ll u=1,ll p=0){
11     for(auto&v:adj[u]){
12         if(v==p)
13             continue;
14         if(d[v]*2>n) return dfs_solve(v,u);
15     }
16     return u;
17 }

```

hld.cpp

```

1  constexpr ll mxN=2e5+1;
2  ll tin=0,in[mxN]={},t[mxN]<<1;
3  ll n,q,a[mxN],d[mxN]={},e[mxN]={};
4  vc<ll> adj[mxN];
5  ll dfs(ll u=1,ll p=0){
6      ll mx=0,sub=1;
7      forn(i,0,adj[u].size()){
8          ll v=adj[u][i];
9          if(v!=p){d[v]=d[u]+1; ll val=dfs(v,u);
10             // sub+=val; if(val>mx) mx=val,e[u]=v;}
11             else swap(adj[u][i],adj[u][0]);
12         }
13     }
14     return sub;
15 }
16 void hld(ll u=1,ll p=1){
17     in[u]++;tin,t[in[u]+n-1]=a[u];
18     if(e[u]) hld(e[u],p);
19     forn(i,1,adj[u].size()){ll v=adj[u][i];
20         // if(e[u]!=v) hld(v,v);}
21         e[u]=p;
22     }
23 }
24 void pst(ll p,ll val){for(t[p+=n-1]=val;p>>=1;t[p]=max(t[p],
25     <<1,t[p<<1|1]));}
26 ll qry(ll l,ll r){
27     ll ans=0;

```

```

23     for(l+=n-1,r+=n-1;l<=r;l>>=1,r>>=1){if(l&1)
24         // ans=max(ans,t[l++]); if(r&1^1)
25         // ans=max(ans,t[r--]);}
26     return ans;
27 }
28 ll query(ll a,ll b){
29     ll ans=0;
30     for(;e[a]!=e[b];b=adj[e[b]][0]){
31         if(d[e[a]]>d[e[b]]) swap(a,b);
32         ans=max(ans,qry(in[e[b]],in[b]));
33     }
34     if(in[a]>in[b]) swap(a,b);
35     return max(ans,qry(in[a],in[b]));
36 }
37 void solve(istream &cin){
38     cin >> n >> q;
39     forn(i,1,n+1) cin >> a[i];
40     forn(i,0,n-1){ll u,v; cin >> u >> v;
41         // adj[u].emp(v),adj[v].emp(u);}
42         adj[u].emp(0);
43         dfs();
44         hld();
45         forr(i,n-1,1) t[i]=max(t[i<<1],t[i<<1|1]);
46         for(ll op,a,b;q--){
47             cin >> op >> a >> b;
48             if(op==1) pst(in[a],b);
49             else cout << query(a,b) << ' ';
50         }
51     }

```

small-to-large.cpp

```

1  // 注意: 每層遞迴持有 deque, 鏈狀樹會爆棧且記憶體~300MB,
2  // 深樹題慎用
3  constexpr ll mxN=2e5+1;
4  ll n,k,res=0;
5  vc<ll> adj[mxN];
6  deque<ll> dfs(ll u=1,ll p=0){
7      deque<ll> cur(1,{1}),tmp;
8      for(auto &v:adj[u]){
9          if(v!=p){
10             tmp=dfs(v,u),tmp.push_front(0);
11             if(tmp.size()>cur.size())
12                 swap(tmp,cur);
13             forn(i,0,tmp.size()) if(k-i>=0 &&
14                 // k-i<cur.size())
15                 // res+=cur[k-i]*tmp[i];
16                 // forn(i,0,tmp.size())
17                 // cur[i]+=tmp[i];
18             }
19         }
20     }
21     return cur;
22 }
23 void solve(istream &cin){
24     cin >> n >> k;
25     forn(i,0,n-1){ll u,v; cin >> u >> v;
26         // adj[u].emp(v),adj[v].emp(u);}
27         dfs();
28         cout << res << '\n';
29     }

```

tree-diameter.cpp

```

1  #define emp emplace_back
2  constexpr ll mxN=2e5+1;
3  ll n,res=0;
4  vc<ll> adj[mxN],dp(mxN);
5
6  void dfs(ll u=1,ll p=0){dp[u]=0; for(auto&v:adj[u])
7      // if(v!=p) dfs(v,u),res=max(res,dp[u]+dp[v]+1),dp[u]=max
8      // (dp[u],dp[v]+1);}
9
10 int main()
11 {

```

```

10     cin>>n; forn(i,0,n-1){ll u,v; cin>>u>>v;
    ↪ adj[u].emp(v),adj[v].emp(u);}
11     dfs();
12     cout << res << '\n';
13 }

```

3 Graph

2-sat.cpp

```

1  constexpr ll mxN=2e5+1;
2  ll sc=0,scc[mxN];
3  ll id=0,st[mxN],vs[mxN]{};
4  ll n,m,tin=0,dfn[mxN]{};low[mxN];
5  vc<ll> adj[mxN];
6  void tarjan(ll u){
7      dfn[u]=low[u]=++tin,st[++id]=u,vs[u]=1;
8      for(auto &v:adj[u]){
9          if(!dfn[v])
    ↪ tarjan(v),low[u]=min(low[u],low[v]);
    else if(vs[v]) low[u]=min(low[u],dfn[v]);
10     }
11     if(low[u]==dfn[u]){
12         sc++;
13         for(;id;){
14             scc[st[id]]=sc,vs[st[id]]=0;
15             id--;
16             if(st[id+1]==u) break;
17         }
18     }
19 }
20 void solve(istream &cin){
21     cin >> n >> m;
22     forn(i,1,n+1){
23         char c1,c2; ll x,y,X,Y; cin >> c1 >> x >>
    ↪ c2 >> y;
24         X=x+m,Y=y+m;
25         if(c1=='-') swap(x,X);
26         if(c2=='-') swap(y,Y);
27         adj[X].emp(y),adj[Y].emp(x);
28     }
29     forn(i,1,2*m+1) if(!dfn[i]) tarjan(i);
30     forn(i,1,m+1) if(scc[i]==scc[i+m]){cout <<
    ↪ "IMPOSSIBLE" << '\n'; return;}
31     forn(i,1,m+1) cout << (scc[i]>scc[i+m]?'-':'+') <<
    ↪ " \n"[i==m];
32 }

```

BellmanFord.cpp

```

1  //SPFA; 回傳false=有負環(僅偵測從start可達的);
    ↪ d[]需外部初始化INF, d[start]=0
2  bool spfa(){
3      vc<ll> vs(n+1,0); //1-indexed 要開 n+1
4      queue<ll> q; q.emplace(start);
5      while(!q.empty()){
6          ll u=q.front(); q.pop(),vs[u]=0;
7          for(auto&[v,w]:adj[u]){
8              if(d[v]>d[u]+w){
9                  d[v]=d[u]+w;
10                 if(!vs[v]){vs[v]=1,cnt[v]++; if(cnt[v]>n)
    ↪ return false; q.emplace(v);}
11             }
12         }
13     }
14     return true;
15 }

```

Dijkstra.cpp

```

1  void djc(){
2      mls<pll> hp;
3      res[1]=0,hp.emplace(pll{res[1],1});
4      while(!hp.empty()){
5          auto [d,u]=*hp.bg(); hp.erase(hp.bg());
6          if(d>res[u]) continue;
7          for(auto&[v,w]:adj[u]){
8              if(w+res[u]<res[v]) res[v]=w+res[u]
    ↪ ,hp.emplace(pll{res[v],v});
9          }
10     }
11 }

```

Edmond.cpp

```

1  constexpr ll mxN=505;
2  ll n,m,adj[mxN][mxN],g[mxN][mxN]{}; //adj=capacity
3  vc<ll> pa;
4  ll bfs(){
5      queue<pll> q; q.emplace(pll{0,inf});
6      while(!q.empty()){
7          auto [u,flow]=q.front(); q.pop();
8          forn(i,0,n)
    ↪ if(g[u][i] && adj[u][i] &&
    ↪ pa[i]==-1){
9              pa[i]=u;
10             if(i==n-1) return
    ↪ min(flow,adj[u][i]);
11             q.emplace(pll{i,min(flow,a
    ↪ dj[u][i])});
12         }
13     }
14     return 0;
15 }
16 void solve(istream &cin){
17     cin>>n>>m;
18     pa.resize(n);
19     forn(i,1,m+1){ll u,v; cin>>u>>v,u--,v--;
    ↪ adj[u][v]++,g[u][v]++,adj[v][u]++,g[v][u]++;}
20     ↪ //重邊容量/重數要累加
21     ll f=0;
22     fill(all(pa),-1),pa[0]=0;
23     for(ll nf;nf=bfs();f+=nf){
24         ll cur=n-1;
25         while(cur) adj[pa[cur]][cur]-=nf,adj[cur][
    ↪ pa[cur]]+=nf,cur=pa[cur];
26         fill(all(pa),-1),pa[0]=0;
27     }
28     vc<pll> res;
29     forn(i,0,n) forn(j,0,n) if(g[i][j] && pa[i]!=-1 &&
    ↪ pa[j]==-1) forn(_,0,g[i][j])
    ↪ res.emplace(pll{i,j}); //重邊每條都要列
30     cout << res.size() << '\n';
31     for(auto&[u,v]:res) cout << u+1 << ' ' << v+1 <<
    ↪ '\n';
32 }
33 int main()
34 {
35     fast_io;
36     ll testcase;
37     // cin>>testcase;
38     testcase=1;
39     while(testcase--){
40         solve(cin);
41     }
42     return 0;
43 }

```

articulation-point.cpp

```

1 //undirected graph articulation point; root呼叫
  ↳ tarjan(1,0), 不連通圖每個分量都要跑
2 ll n,tim=0,art=0;
3 vc<ll> dfn(mxN,0),low(mxN);
4 void tarjan(ll u,ll p){
5     ll cnt=0; bool isArt=false;
6     dfn[u]=low[u]=++tim;
7     for(auto&v:adj[u]){
8         if(!dfn[v]){
9             tarjan(v,u);
10            low[u]=min(low[u],low[v]);
11            if(low[v]>=dfn[u] && p) isArt=true; //非root:
  ↳ 某子樹回不到u之上
12            cnt++;
13        }
14        else if(v!=p) low[u]=min(low[u],dfn[v]);
15    }
16    if(!p && cnt>1) isArt=true; //root: 至少兩個子樹
17    art+=isArt;
18 }

```

bridge.cpp

```

1 constexpr ll mxN=1e5+1;
2 ll n,m,dfn[mxN]={},low[mxN],tin=0;
3 vc<ll> adj[mxN];
4 vc<ar<ll,2>> res;
5 void tarjan(ll u=1,ll p=0){
6     dfn[u]=low[u]=++tin;
7     bool skip=0; //父邊只跳過一次,
  ↳ 重邊(平行邊)才不會被誤判成橋
8     for(auto &v:adj[u]){
9         if(v==p && !skip){skip=1; continue;}
10        if(dfn[v]) low[u]=min(low[u],dfn[v]);
11        else{
12            tarjan(v,u),low[u]=min(low[u],low
13            ↳ v)];
14            if(low[v]>dfn[u])
15            ↳ res.emp(ar<ll,2>{u,v});
16        }
17    }
18 }

```

dinic.cpp

```

1
2 constexpr ll mxN=505;
3 class fe{
4     public:
5         ll u,v,cp,fw;
6         fe(){
7             fe(ll U,ll V,ll CP):u(U),v(V),cp(CP){fw=0;}
8         };
9         ll n,m,sz=0;
10        vc<fe> edg;
11        queue<ll> q;
12        vc<ll> adj[mxN],ptr,dep;
13        bool bfs(){
14            while(!q.empty()){
15                ll u=q.front(); q.pop();
16                for(auto&id:adj[u]){
17                    if(edg[id].cp==edg[id].fw ||
18                    ↳ dep[edg[id].v]!=-1) continue;
19                    dep[edg[id].v]=dep[u]+1,q.emplace(
20                    ↳ edg[id].v);
21                }
22            }
23            return dep[n-1]!=-1;
24        }
25        ll dinic(ll u,ll push){
26            if(!push) return 0;

```

```

27        if(u==n-1) return push;
28        for(ll &cid=ptr[u];cid<adj[u].size();++cid){
29            ll id=adj[u][cid];
30            ll v=edg[id].v;
31            if(dep[v]!=dep[u]+1) continue;
32            ll tr=dinic(v,min(push,edg[id].cp-edg[id].
33            ↳ fw));
34            if(!tr) continue;
35            edg[id].fw+=tr,edg[id^1].fw-=tr;
36            return tr;
37        }
38        return 0;
39    }
40    void solve(istream &cin){
41        cin>>n>m;
42        ptr.resize(n),dep.resize(n);
43        forn(i,1,m+1){
44            ll u,v,c; cin>>u>>v>>c,u--,v--;
45            edg.emp(fe{u,v,c}),edg.emp(fe{v,u,0});
46            adj[u].emp(sz),adj[v].emp(sz+1),sz+=2;
47        }
48        ll res=0;
49        for(;;){
50            ll flow;
51            fill(all(dep),-1);
52            q.emplace(0),dep[0]=0;
53            if(!bfs()) break;
54            fill(all(ptr),0);
55            for(;;flow=dinic(0,inf);res+=flow);
56        }
57        cout << res << '\n';
58    }
59    int main()
60    {
61        fast_io;
62        ll testcase;
63        // cin>>testcase;
64        testcase=1;
65        while(testcase--){
66            solve(cin);
67            return 0;
68        }
69    }

```

dominator-tree.cpp

```

1 constexpr ll mxN=1e5+1;
2 ll n,m,tim=0,id[mxN],sd[mxN],pa[mxN],f[mxN],lb[mxN],et[mxN]
  ↳ []{},ret[mxN];
3 vc<ll> adj[mxN],radj[mxN],t[mxN],st;
4 void dfs(ll u=1){
5     et[u]=++tim,ret[tim]=u;
6     id[tim]=sd[tim]=lb[tim]=pa[tim]=tim;
7     for(auto&v:adj[u]){
8         if(!et[v]) dfs(v),f[et[v]]=et[u];
9         radj[et[v]].emp(et[u]);
10    }
11    ll find(ll u,ll x=0){
12        if(u==pa[u]) return x?-1:u;
13        ll v=find(pa[u],x+1);
14        if(v<0) return u;
15        if(sd[lb[u]]>sd[lb[pa[u]]]) lb[u]=lb[pa[u]];
16        pa[u]=v;
17        return x?v:lb[u];
18    }
19    void built(){
20        vc<ll> arr[n+1];
21        forr(i,n,1){
22            for(auto&v:radj[i])
23            ↳ sd[i]=min(sd[i],sd[find(v)]);
24            if(i>1) arr[sd[i]].emp(i);
25            for(auto&v:arr[i]){
26                ll x=find(v);
27                if(sd[x]==sd[v]) id[v]=sd[v];
28                else id[v]=x;
29            }

```

```

30         if(i>1) pa[i]=f[i];
31     }
32     forn(i,2,n+1){
33         if(id[i]!=sd[i]) id[i]=id[id[i]];
34         t[ret[i]].emp(ret[id[i]]),t[ret[id[i]]].emp
            ↪ p(ret[i]);
35     }
36 }
37 void dfs_ans(ll u=1,ll p=0){
38     st.emp(u);
39     if(u==n){
40         sort(all(st));
41         cout << st.size() << '\n';
42         for(auto&v:st) cout << v << ' '; cout <<
            ↪ '\n';
43         exit(0);
44     }
45     for(auto&v:t[u]){
46         if(v==p) continue;
47         dfs_ans(v,u);
48     }
49     st.pop_back();
50 }
51 void solve(istream &cin){
52     cin>>n>>m;
53     forn(i,1,m+1){
54         ll u,v; cin>>u>>v;
55         adj[u].emp(v);
56     }
57     dfs();
58     built();
59     dfs_ans();
60 }
61

```

floyd-warshall.cpp

```

1 // 先設 f[x][x]=0 其餘INF
   ↪ (INF用0x3f3f3f3f3f3f3f3f可安全相加)
2 for (ll k = 1; k <= n; k++) {
3     for (ll x = 1; x <= n; x++) {
4         for (ll y = 1; y <= n; y++) {
5             f[x][y] = min(f[x][y], f[x][k] + f[k][y]);
6         }
7     }
8 }

```

hierholzer.cpp

```

1 //debrujin sequence
2 ll n;
3 string s;
4 vc<ll> adj[1<<14|1];
5 void dfs(ll u=0){
6     while(!adj[u].empty()){
7         ll v; v=adj[u].back(); adj[u].pop_back();
8         dfs(v);
9         s+=(v&1?'1':'0');
10    }
11 }
12 void solve(istream &cin){
13     cin>>n;
14     if(n==1){cout << "01" << '\n'; return;}
15     forn(i,0,1<<(n-1)){
16         adj[i].emp((i<<1)&((1<<(n-1))-1));
17         adj[i].emp((i<<1|1)&((1<<(n-1))-1));
18     }
19     dfs();
20     forn(i,0,n-1) s+='0';
21     reverse(all(s));
22     cout << s << '\n';
23 }

```

hopcroft-karp.cpp

```

1 constexpr ll mxN=501;
2 ll n,m,k,mat[mxN<<1]{},d[mxN<<1];
3 vc<ll> adj[mxN];
4 ll bfs(){
5     queue<ll> q;
6     memset(d,-1,sizeof(d));
7     forn(i,1,n+1) if(!mat[i]) d[i]=0,q.emplace(i);
8     while(!q.empty()){
9         ll u=q.front(); q.pop();
10        if(~d[0]) return 1;
11        for(auto &v:adj[u]){
12            if(!~d[mat[v]]) d[mat[v]]=d[u]+1
            ↪ ,q.emplace(mat[v]);
13        }
14    }
15    return 0;
16 }
17 bool dfs(ll u){
18     if(!u) return true;
19     for(auto &v:adj[u]) if(d[mat[v]]==d[u]+1 &&
        ↪ dfs(mat[v])){mat[u]=v,mat[v]=u;return true;}
20     d[u]=-1;
21     return false;
22 }
23 void solve(istream &cin){
24     cin >> n >> m >> k;
25     forn(i,0,k){ll u,v; cin >> u >> v;
        ↪ adj[u].emp(n+v);}
26     ll res=0;
27     for(;bfs();){forn(i,1,n+1) if(!mat[i]) res+=dfs(i);
28     cout << res << '\n';
29     if(res) forn(i,1,n+1) if(mat[i]) cout << i << ' '
        ↪ << (mat[i]-n) << '\n';
30 }

```

kuhn.cpp

```

1 constexpr ll mxN=1001;
2 constexpr ll base=500;
3 ll n,m,k,vs1[mxN],vs2[mxN],ok[mxN][mxN]{};
4 vc<ll> adj[mxN];
5 bool dfs(ll u){
6     if(vs2[u]) return false;
7     vs2[u]=1;
8     for(auto&v:adj[u]) if(!vs1[v] ||
        ↪ dfs(vs1[v])){vs1[v]=u; return true;}
9     return false;
10 }
11 void solve(istream &cin){
12     cin >> n >> m >> k;
13     //[1,500] boy [500+1,500+500] girl
14     forn(i,0,k){
15         ll u,v; cin >> u >> v,v+=base;
16         if(!ok[u][v])
17             adj[u].emp(v),adj[v].emp(u),ok[u][v]=1;
18     }
19     memset(vs1,0,sizeof(vs1));
20     forn(i,1,n+m+1){
21         memset(vs2,0,sizeof(vs2));
22         dfs(i);
23     }
24     vc<ar<ll,2>> res;
25     forn(i,1,n+1){
26         for(auto &v:adj[i]){
27             if(vs1[v]==i)
                ↪ res.emp(ar<ll,2>{i,v-base});
28         }
29     }
30     cout << res.size() << '\n';
31     for(auto&[u,v]:res) cout << u << ' ' << v << '\n';
32 }

```


mcmf.cpp

```

1  class Edge{
2      public:
3          ll u,v,c,w;
4          Edge(){
5              Edge(ll u,ll v,ll c,ll
6                  ↪ w):u(u),v(v),c(c),w(w){}
7  };
8  constexpr ll mxN=501;
9  ll n,m,k,id=0,vs[mxN],d[mxN],pa[mxN];
10 vc<Edge> e;
11 vc<ll> adj[mxN];
12 ll bfs(){
13     memset(pa,-1,8*(n+1)),memset(d,0x3f,sizeof(d));
14     queue<ll> q;
15     d[1]=0,vs[1]=1,q.emplace(1);
16     while(!q.empty()){
17         auto u=q.front(); q.pop(),vs[u]=0;
18         for(auto &eid:adj[u]){
19             auto [x,v,c,w]=e[eid];
20             if(c && d[u]+w<d[v]){
21                 pa[v]=eid,d[v]=d[u]+w;
22                 if(!vs[v])
23                     ↪ vs[v]=1,q.emplace(v);
24             }
25         }
26     }
27     return d[n]<INF;
28 }
29 void solve(){
30     cin >> n >> m >> k;
31     forn(i,1,m+1){
32         ll u,v,c,w; cin >> u >> v >> c >> w;
33         e.emp(Edge{u,v,c,w}),e.emp(Edge{v,u,0,-w});
34         ↪ ,adj[u].emp(id++),adj[v].emp(id++);
35     }
36     ll f=0,res=0;
37     for(ll nf,v;f<k && bfs();){
38         for(nf=k-f,v=n;v!=1;nf=min(nf,e[pa[v]].c),
39             ↪ v=e[pa[v]].u);
40         for(v=n;v!=1;e[pa[v]].c-=nf,e[pa[v]^1].c+=
41             ↪ nf,v=e[pa[v]].u);
42         res+=d[n]*nf,f+=nf;
43     }
44     cout << (f<k?-1:res) << '\n';
45 }

```

scc.cpp

```

1  void dfs(ll u){
2      dfn[u]=low[u]=++tim,st[++id]=u,vs[u]=1;
3      for(auto &v:adj[u]){
4          if(!dfn[v])
5              ↪ dfs(v),low[u]=min(low[u],low[v]);
6          else if(vs[v]) low[u]=min(low[u],dfn[v]);
7      }
8      if(low[u]==dfn[u]){
9          for(ll v=-1;v!=u;scc[st[id]]=sc,vs[st[id]]
10             ↪ =0,v=st[id],id--);
11          sc++;
12      }
13 }

```

toposort.cpp

```

1  void topo_sort() {
2      int n, m;
3      cin >> n >> m;
4      vector<vector<int>> graph(n+1);
5      for (int i = 0; i < m; i++) {
6          int a, b;
7          cin >> a >> b;

```

```

graph[a].push_back(b);
}
vector<int> rdu(n+1, 0);
for (auto &node : graph) {
    for (auto &now : node) {
        rdu[now]++;
    }
}
queue<int> que;
for (int i = 1; i <= n; i++) {
    if (rdu[i] == 0) que.push(i);
}
vector<int> topo;
while (!que.empty()) {
    int top = que.front();
    que.pop();
    topo.push_back(top);
    for (auto &i : graph[top]) {
        rdu[i]--;
        if (rdu[i] == 0) que.push(i);
    }
}
if (topo.size() == n) {
    for (auto &i : topo) cout << i << " ";
}
else
    cout << "IMPOSSIBLE";
}

```

4 Geometry

Determine whether a polygon is convex.cpp

```

// 注意:
↪ 自交但每個轉向同號的多邊形(如五角星一筆畫)會被誤判convex,
// 嚴格判定需另檢查總轉角恰為±360度; 整數座標>~3e7時double
↪ cross有精度風險, 應改ll
#include <vector>
using namespace std;

struct Point {
    double x, y;
};

double cross(const Point& a, const Point& b, const Point&
    ↪ c) {
    // 計算向量 ab x bc
    double x1 = b.x - a.x, y1 = b.y - a.y;
    double x2 = c.x - b.x, y2 = c.y - b.y;
    return x1 * y2 - y1 * x2;
}

bool isConvex(const vector<Point>& poly) {
    int n = poly.size();
    if (n < 3) return false;
    int sign = 0;
    for (int i = 0; i < n; ++i) {
        double c = cross(poly[i], poly[(i + 1) % n],
            ↪ poly[(i + 2) % n]);
        if (c != 0) {
            if (sign == 0)
                sign = (c > 0) ? 1 : -1;
            else if ((c > 0 && sign < 0) || (c < 0 && sign
                ↪ > 0))
                return false;
        }
    }
    return true;
}

```


Rotating Calipers 求凸包直徑.cpp

```

1  struct Point {
2      double x, y;
3  };
4
5  double dist2(const Point& a, const Point& b) {
6      double dx = a.x - b.x, dy = a.y - b.y;
7      return dx * dx + dy * dy;
8  }
9
10 // 假設 convex 是逆時針的「嚴格」凸包(無共線點/重複點,
    ↳ 否則 two-pointer 不保證正確)
11 // 座標 > 1e7 時 double 平方和超過 2^53 有精度風險,
    ↳ 應改 ll 算 dist2 最後才 sqrt
12 double convexHullDiameter(const vector<Point>& convex) {
13     int n = convex.size();
14     if (n == 1) return 0;
15     if (n == 2) return sqrt(dist2(convex[0], convex[1]));
16     double res = 0;
17     for (int i = 0, j = 1; i < n; ++i) {
18         while (dist2(convex[i], convex[(j + 1) % n]) >
            ↳ dist2(convex[i], convex[j]))
19             j = (j + 1) % n;
20         res = max(res, dist2(convex[i], convex[j]));
21     }
22     return sqrt(res);
23 }

```

all solutions for Diophantine Equation in given range.cpp

```

1  /*
2      1. General solution to the linear Diophantine equation:
3          a * x + b * y = c
4          If (x0, y0) is a particular solution and g = gcd(a,
            ↳ b):
5              x = x0 + k * (b / g)
6              y = y0 - k * (a / g)
7              for any integer k
8          2. Number of integer solutions (x, y) within bounds:
9              minx <= x <= maxx
10             miny <= y <= maxy
11             需 a != 0 且 b != 0; |a|, |b|, |c|, 範圍 ~1e9 內不溢位
12 */
13 ll gcd_ext(ll a, ll b, ll &x, ll &y) {
14     if (!b) { x = 1, y = 0; return a; }
15     ll x1, y1, g = gcd_ext(b, a % b, x1, y1);
16     x = y1, y = x1 - (a / b) * y1;
17     return g;
18 }
19 bool find_any_solution(ll a, ll b, ll c, ll &x0, ll &y0,
    ↳ ll &g) {
20     g = gcd_ext(abs(a), abs(b), x0, y0);
21     if (c % g) return false;
22     x0 *= c / g, y0 *= c / g;
23     if (a < 0) x0 = -x0;
24     if (b < 0) y0 = -y0;
25     return true;
26 }
27 void shift_solution(ll &x, ll &y, ll a, ll b, ll cnt) {
28     x += cnt * b;
29     y -= cnt * a;
30 }
31
32 ll find_all_solutions(ll a, ll b, ll c, ll minx, ll maxx,
    ↳ ll miny, ll maxy) {
33     ll x, y, g;
34     if (!find_any_solution(a, b, c, x, y, g))
35         return 0;
36     a /= g;
37     b /= g;
38
39     ll sign_a = a > 0 ? +1 : -1;
40     ll sign_b = b > 0 ? +1 : -1;

```

```

41 shift_solution(x, y, a, b, (minx - x) / b);
42
43 if (x < minx)
44     shift_solution(x, y, a, b, sign_b);
45 if (x > maxx)
46     return 0;
47 ll lx1 = x;
48
49 shift_solution(x, y, a, b, (maxx - x) / b);
50 if (x > maxx)
51     shift_solution(x, y, a, b, -sign_b);
52 ll rx1 = x;
53
54 shift_solution(x, y, a, b, -(miny - y) / a);
55 if (y < miny)
56     shift_solution(x, y, a, b, -sign_a);
57 if (y > maxy)
58     return 0;
59 ll lx2 = x;
60
61 shift_solution(x, y, a, b, -(maxy - y) / a);
62 if (y > maxy)
63     shift_solution(x, y, a, b, sign_a);
64 ll rx2 = x;
65
66 if (lx2 > rx2)
67     swap(lx2, rx2);
68 ll lx = max(lx1, lx2);
69 ll rx = min(rx1, rx2);
70
71 if (lx > rx)
72     return 0;
73 return (rx - lx) / abs(b) + 1;
74 }

```

convex-hull.cpp

```

1 // 注意: 重複點會全部留在輸出裡, 用前 sort+unique 去重
2 // 共線邊上的點會保留(非嚴格凸包); /座標 <= 1e9 時 cross
    ↳ 不溢位
3 class Point {
4     public:
5         ll x, y;
6         Point() {}
7         Point(ll x, ll y): x(x), y(y) {}
8         friend ll operator^(const Point a, const
            ↳ Point b){return a.x*b.y - a.y*b.x;}
9         friend ll operator*(const Point a, const
            ↳ Point b){return a.x*b.x + a.y*b.y;}
10        friend Point operator-(const Point a, const
            ↳ Point b){
11            return Point{a.x-b.x, a.y-b.y};
12        }
13        friend ostream& operator>>(ostream
            ↳ &in, Point &a){
14            in >> a.x >> a.y;
15            return in;
16        }
17        friend ostream& operator<<(ostream
            ↳ &out, Point &a){
18            out << a.x << ' ' << a.y << '\n';
19            return out;
20        }
21        friend bool operator<(const Point a, const
            ↳ Point b){return a.x < b.x || (a.x == b.x &&
            ↳ a.y < b.y);}
22    };
23 ll dis(Point a, Point b){
24     return (a.x-b.x)*(a.x-b.x) + (a.y-b.y)*(a.y-b.y);
25 }
26 ll ok(Point a, Point b, Point c){
27     ll ang = (b-a)^(c-a); //ang>0 false ang< true
28     return ang < 0;
29 }
30 constexpr ll mxN = 2e5+5;
31 ll st[mxN], id = 0;
32 void solve(istream &cin){

```

```

33     ll n; cin >> n;
34     vc<Point> a(n); for(auto &v:a) cin >> v;
35     forn(i,1,n) if(a[i]<a[0]) swap(a[i],a[0]);
36     sort(1+all(a),[p=a[0]](Point a,Point b){ll
        ↪ ang=(a-p)^(b-p); return
        ↪ !ang?dis(p,a)<dis(p,b):ang>0;});
37     {
38         ll j; for(j=n-1;j>=0 &&
        ↪ ((a[j]-a[0])^(a[n-1]-a[0]))==0;j--);
        ↪ reverse(j+1+a.bg(),a.ed());
39     }
40     forn(i,0,n){
41         while(id>1 &&
        ↪ ok(a[st[id-1]],a[st[id]],a[i])) --id;
42         st[++id]=i;
43     }
44     cout << id << '\n';
45     if(id) for(;id;cout << a[st[id--]]);
46 }

```

cross-product.cpp

```

1  class Point{
2      public:
3          ll x,y;
4          Point(){
5              Point(ll x,ll y):x(x),y(y){
6                  ll f(Point a,Point b){
7                      return a.x*b.y-b.x*a.y;
8                  }//cross product
9                  friend Point operator-(const Point a,const
        ↪ Point b){
10                     return Point{a.x-b.x,a.y-b.y};
11                 }
12                 friend istream& operator>>(istream
        ↪ &in,Point &a){
13                     in >> a.x >> a.y;
14                     return in;
15                 }
16 };

```

extend-li-chao-tree.cpp

```

1  // 注意：空節點預設 Line(0,-1) 是此題專用sentinel,
    ↪ 答案可能<-1的題要改-inf
2  // solve()斜率 (y2-y1)/(x2-x1)：垂直線段(x1==x2)會除以零，
    ↪ 不整除會截斷
3  class Line{
4      public:
5          ll m,c;
6          Line(){m=0,c=-1;}
7          Line(ll m,ll c):m(m),c(c){
8              ll cal(ll x){return m*x+c;}
9          };
10     constexpr ll mxN=1e5;
11     Line t[mxN<<2];
12     void insert(ll i,ll sl,ll sr,ll l,ll r,Line L){
13         if(l>sr || r<sl) return ;
14         ll mid=(sl+sr)>>1;
15         if(l<=sl && sr<=r){
16             if(L.cal(mid)>t[i].cal(mid)) swap(L,t[i]);
17             if(sl==sr) return;
18             if(L.m>t[i].m)
        ↪ insert(i<<1|1,mid+1,sr,l,r,L);
        ↪ else insert(i<<1,sl,mid,l,r,L);
        ↪ return ;
19         }
20         insert(i<<1,sl,mid,l,r,L);
21         insert(i<<1|1,mid+1,sr,l,r,L);
22     }
23     ll query(ll i,ll sl,ll sr,ll x){
24         ll mid=(sl+sr)>>1;
25         if(sl==sr) return t[i].cal(x);
26         if(l>sr || r<sl) return ;
27         if(l<=sl && sr<=r){
28             if(L.cal(mid)>t[i].cal(mid)) swap(L,t[i]);
29             if(sl==sr) return;
30             if(L.m>t[i].m)
        ↪ insert(i<<1|1,mid+1,sr,l,r,L);
        ↪ else insert(i<<1,sl,mid,l,r,L);
        ↪ return ;
31         }
32         insert(i<<1,sl,mid,l,r,L);
33         insert(i<<1|1,mid+1,sr,l,r,L);
34     }
35     ll query(ll i,ll sl,ll sr,ll x){
36         ll mid=(sl+sr)>>1;
37         if(sl==sr) return t[i].cal(x);
38         if(l>sr || r<sl) return ;
39         if(l<=sl && sr<=r){
40             if(L.cal(mid)>t[i].cal(mid)) swap(L,t[i]);
41             if(sl==sr) return;
42             if(L.m>t[i].m)
        ↪ insert(i<<1|1,mid+1,sr,l,r,L);
        ↪ else insert(i<<1,sl,mid,l,r,L);
        ↪ return ;
43         }
44         insert(i<<1,sl,mid,l,r,L);
45         insert(i<<1|1,mid+1,sr,l,r,L);
46     }

```

```

28     if(x<=mid) return
        ↪ max(t[i].cal(x),query(i<<1,sl,mid,x));
29     else return
        ↪ max(t[i].cal(x),query(i<<1|1,mid+1,sr,x));
30 }
31 void solve(istream &cin){
32     ll n,m,r0=-1; cin >> n >> m;
33     forn(i,0,n){
34         ll x1,x2,y1,y2; cin >> x1 >> y1 >> x2 >>
        ↪ y2;
35         if(!x1) r0=max(r0,y1);
36         Line l((y2-y1)/(x2-x1),-x1*(y2-y1)/(x2-x1)
        ↪ + y1);
        insert(1,1,mxN,x1,x2,1);
37     }
38     cout << r0 << " \n"[m==1];
39     forn(i,1,m+1) cout << query(1,1,mxN,i) << "
        ↪ \n"[i==m];
40 }

```

li-chao-tree.cpp

```

1  // 注意：空節點預設 Line(0,0)，查詢取 max 所以答案被鎖在 >=0;
2  // 答案可能為負的題要改 Line(){m=0,c=-inf;}
    ↪ (m=0所以cal不會溢位)
3  // solve()是特定題：(y2-y1)/m 整數除法，斜率不整除會錯，
    ↪ 通用時只抄struct+insert+query
4  class Line{
5      public:
6          ll m,c;
7          Line(){m=c=0;}
8          Line(ll m,ll c):m(m),c(c){
9              ll cal(ll x){return m*x+c;}
10     };
11     constexpr ll mxN=1e5;
12     Line t[mxN<<2];
13     void insert(ll i,ll sl,ll sr,Line l){
14         ll mid=(sl+sr)>>1;
15         if(l.cal(mid)>t[i].cal(mid)) swap(l,t[i]);
16         if(sl==sr) return;
17         if(l.m>t[i].m) insert(i<<1|1,mid+1,sr,l);
18         else insert(i<<1,sl,mid,l);
19     }
20     ll query(ll i,ll sl,ll sr,ll x){
21         ll mid=(sl+sr)>>1;
22         if(sl==sr) return t[i].cal(x);
23         if(x<=mid) return
        ↪ max(t[i].cal(x),query(i<<1,sl,mid,x));
        ↪ else return
        ↪ max(t[i].cal(x),query(i<<1|1,mid+1,sr,x));
24     }
25 }
26 void solve(istream &cin){
27     ll n,m; cin >> n >> m;
28     forn(i,0,n){
29         ll y1,y2; cin >> y1 >> y2;
30         // (y2-y1)/(m-0) = (y-y1)/x
        ↪ //x * (y2-y1)/m + y1 = y
        Line l((y2-y1)/m,y1);
        insert(1,1,mxN,1);
31     }
32     forn(i,0,m+1) cout << query(1,1,mxN,i) << "
        ↪ \n"[i==m];
33 }

```

max-man-dis.cpp

```

1  constexpr ll mxN=2e5+1;
2  ll n,d[2][2];
3  void solve(){
4      cin >> n;
5      d[0][0]=d[1][0]=-inf;
6      d[0][1]=d[1][1]=inf;
7      forn(i,0,n){
8          ll x,y;

```

```

9         cin >> x >> y;
10        d[0][0]=max(d[0][0],x+y);
11        d[0][1]=min(d[0][1],x+y);
12        d[1][0]=max(d[1][0],x-y);
13        d[1][1]=min(d[1][1],x-y);
14        cout <<
        ↪ max(d[0][0]-d[0][1],d[1][0]-d[1][1])
        ↪ << '\n';
15    }
16 }

```

minimum-euc-dis.cpp

```

1  class Point{
2      public:
3          ll x,y;
4          Point(){ }
5          Point(ll x,ll y):x(x),y(y){ }
6          friend ll operator^(const Point a,const
        ↪ Point b){return a.x*b.y-a.y*b.x;}
7          friend ll operator*(const Point a,const
        ↪ Point b){return a.x*b.x+a.y*b.y;}
8          friend Point operator-(const Point a,const
        ↪ Point b){
9              return Point{a.x-b.x,a.y-b.y};
10         }
11         friend istream& operator>>(istream
        ↪ &in,Point &a){
12             in >> a.x >> a.y;
13             return in;
14         }
15         friend bool operator<(const Point a,const
        ↪ Point b){return a.x<b.x||(a.x==b.x &&
        ↪ a.y<b.y);}
16 };
17 ll euc(Point a,Point b){
18     return (a.x-b.x)*(a.x-b.x)+(a.y-b.y)*(a.y-b.y);
19 }
20 void solve(istream &cin){
21     ll n; cin >> n;
22     vc<Point> a(n); for(auto &v:a) cin >> v;
23     sort(all(a),[](Point u,Point v){return
        ↪ (u.y<v.y)|| (u.y==v.y && u.x<v.x)});
24     ll d;
25     mls<Point> hp;
26     d=euc(a[0],a[1]),hp.emplace(a[0]),hp.emplace(a[1]);
27     for(ll i=2,p=0;i<n;i++){
28         for(;p<i && (a[p].y-a[i].y)*(a[p].y-a[i].y)
        ↪ )>=d;hp.erase(hp.lwb(a[p])),p++);
29         ↪ //p<i! 寫p<n重複點(d=0)會清空set後eras
        ↪ (end())炸掉
30         auto it=hp.lwb(a[i]);
31         auto front=it,back=it;
32         forn(j,0,4){
33             if(front!=hp.ed())
        ↪ d=min(d,euc(a[i],*front));
34             else break; front++;
35         }
36         forn(j,0,4){
37             if(back!=hp.ed())
        ↪ d=min(d,euc(a[i],*back));
38             {if(back!=hp.bg()) back--; else
        ↪ break;}
39         }
40         hp.emplace(a[i]);
41     }
42     cout << d << '\n';

```

pick-theorem.cpp

```

1  void solve(istream &cin){
2      ll n; cin >> n;
3      vc<Point> a(n);

```

```

4      for(auto &v:a) cin >> v;
5      ll area=0,B=0;
6      forn(i,0,n){
7          ll j=(i+1)%n;
8          area+=a[i]^a[j];
9          B+=gcd(abs(a[i].x-a[j].x),abs(a[i].y-a[j].y));
10     }
11     cout << (abs(area)/2+1-B/2) << ' ' << B << '\n';
12     //area=I+B/2-1 I=internal node B=boundary node
13 }

```

ray-casting.cpp

```

1  class Point{
2      public:
3          ll x,y;
4          Point(){ }
5          Point(ll x,ll y):x(x),y(y){ }
6          friend ll operator^(const Point a,const
        ↪ Point b){return a.x*b.y-a.y*b.x;}
7          friend ll operator*(const Point a,const
        ↪ Point b){return a.x*b.x+a.y*b.y;}
8          friend Point operator-(const Point a,const
        ↪ Point b){
9              return Point{a.x-b.x,a.y-b.y};
10         }
11         friend istream& operator>>(istream
        ↪ &in,Point &a){
12             in >> a.x >> a.y;
13             return in;
14         }
15     };
16     bool on(Point &a,Point &b,Point &c){
17         if((b-a)^(c-a)) return false;
18         return 0<=((b-a)*(c-a)) &&
        ↪ ((b-a)*(c-a))<=((b-a)*(b-a));
19     }
20     void query(vc<Point> &a,Point &p){
21         ll n=a.size(),ok=0;
22         forn(i,0,n){
23             ll j=(i+1)%n;
24             if(on(a[i],a[j],p)){cout << "BOUNDARY" <<
        ↪ '\n'; return;}
25             if(p.y>a[i].y!=p.y>a[j].y){
26                 ll sg=(a[j]-a[i])^(p-a[i]);
27                 if(sg<0==a[i].y>a[j].y) ok^=1;
28             }
29         }
30         cout << (ok&1?"INSIDE":"OUTSIDE") << '\n';
31     }
32 }

```

動態凸包.cpp

```

1  struct Point {
2      long long x, y;
3      bool operator<(const Point& p) const {
4          return x < p.x || (x == p.x && y < p.y);
5      }
6      bool operator==(const Point& p) const { return x ==
        ↪ p.x && y == p.y; }
7  };
8
9  long long cross(const Point& a, const Point& b, const
        ↪ Point& c) {
10     // (b-a) x (c-a)
11     return (b.x - a.x) * (c.y - a.y) - (b.y - a.y) * (c.x
        ↪ - a.x);
12 }
13
14 // 只支援插入; 嚴格凸包 (共線點會被移除), 每條殼同 x 只留一點
15 struct DynamicConvexHull {
16     set<Point> up, dn; // 上凸殼、下凸殼 (皆由左至右)

```

```

17 void insert(const Point& p) {
18     insertHull(up, p, +1);
19     insertHull(dn, p, -1);
20 }
21
22 // 逆時針凸包頂點 (不重複); 1 點/2 點退化也正確
23 vector<Point> getConvexHull() const {
24     vector<Point> res(dn.begin(), dn.end());
25     for (auto it = up.rbegin(); it != up.rend(); ++it)
26         if (!res.size() || *it == res.back() || *it
27             == res.front())
28             res.push_back(*it);
29     return res;
30 }
31
32 private:
33 // side=+1 上殼, -1 下殼; p 在殼上或內側回傳 true
34 bool covered(const set<Point>& h, const Point& p, long
35     long side) const {
36     if (h.empty() || p.x < h.begin()->x || p.x >
37         h.rbegin()->x) return false;
38     auto r = h.lower_bound(Point{p.x, LLONG_MIN}); //
39     if (r->x == p.x) return side * (r->y - p.y) >= 0;
40     return side * cross(*prev(r), *r, p) <= 0;
41 }
42 void insertHull(set<Point>& h, const Point& p, long
43     long side) {
44     if (covered(h, p, side)) return; // 在殼內,
45     // 不影響凸包
46     auto r = h.lower_bound(Point{p.x, LLONG_MIN});
47     if (r != h.end() && r->x == p.x) h.erase(r); //
48     // 同x被p支配
49     auto it = h.insert(p).first;
50     while (it != h.begin()) { // 往左刪不再凸的點
51         auto b = prev(it);
52         if (b == h.begin()) break;
53         auto a = prev(b);
54         if (side * cross(*a, *b, *it) >= 0) h.erase(b);
55         else break;
56     }
57     while (next(it) != h.end()) { // 往右刪不再凸的點
58         auto b = next(it);
59         if (next(b) == h.end()) break;
60         auto c = next(b);
61         if (side * cross(*it, *b, *c) >= 0) h.erase(b);
62         else break;
63     }
64 }
65 };

```

多邊形重心計算.cpp

```

1 // 注意: 面積為0(退化/全共線)會除以零得NaN;
2 // 整數座標~3e7時double精度風險(可改ll累加)
3 struct Point {
4     double x, y;
5 };
6 Point polygonCentroid(const vector<Point>& poly) {
7     double cx = 0, cy = 0, area = 0;
8     int n = poly.size();
9     for (int i = 0; i < n; ++i) {
10         double cross = poly[i].x * poly[(i + 1) % n].y -
11             poly[(i + 1) % n].x * poly[i].y;
12         cx += (poly[i].x + poly[(i + 1) % n].x) * cross;
13         cy += (poly[i].y + poly[(i + 1) % n].y) * cross;
14         area += cross;
15     }
16     area /= 2.0;
17     cx /= (6.0 * area);
18     cy /= (6.0 * area);
19     return {cx, cy};
20 }

```

找三點組成最大圓.cpp

```

1 // !!! 已驗證此演算法是錯的: 「最大外接圓的三點在凸包上」
2 // 的前提不成立,
3 // !!! 隨機測資 2/3 輸出錯誤(誤差可達10倍)。正解基本上只能
4 // 0(n^3) 枚舉。勿抄!
5 #include <bits/stdc++.h>
6 using namespace std;
7 typedef long long ll;
8 typedef pair<ll, ll> P;
9
10 ll cross(P a, P b) { return a.first * b.second - a.second
11     * b.first; }
12 ll dist2(P a, P b) {
13     ll dx = a.first - b.first, dy = a.second - b.second;
14     return dx * dx + dy * dy;
15 }
16 double dist(P a, P b) { return sqrt(dist2(a, b)); }
17
18 // Andrew's Monotone Chain
19 vector<P> convex_hull(vector<P> pts) {
20     sort(pts.begin(), pts.end());
21     int n = pts.size(), k = 0;
22     vector<P> hull(2 * n);
23     for (int i = 0; i < n; ++i) {
24         while (k >= 2 &&
25             cross({hull[k-1].first-hull[k-2].first,
26                 hull[k-1].second-hull[k-2].second},
27                 {pts[i].first-hull[k-1].first,
28                     pts[i].second-hull[k-1].second}) <=
29                     0)
30             --k;
31         hull[k++] = pts[i];
32     }
33     for (int i = n-2, t = k; i >= 0; --i) {
34         while (k > t &&
35             cross({hull[k-1].first-hull[k-2].first,
36                 hull[k-1].second-hull[k-2].second},
37                 {pts[i].first-hull[k-1].first,
38                     pts[i].second-hull[k-1].second}) <=
39                     0)
40             --k;
41         hull[k++] = pts[i];
42     }
43     hull.resize(k-1);
44     return hull;
45 }
46
47 double circumradius(P a, P b, P c) {
48     double A = dist(b, c), B = dist(a, c), C = dist(a, b);
49     double S = abs(cross({b.first-a.first,
50         b.second-a.second},
51         {c.first-a.first,
52             c.second-a.second})) / 2.0;
53     if (S == 0) return 0; // 共線
54     return (A * B * C) / (4.0 * S);
55 }
56
57 int main() {
58     int n;
59     cin >> n;
60     vector<P> pts(n);
61     for (int i = 0; i < n; ++i) cin >> pts[i].first >>
62         pts[i].second;
63     vector<P> hull = convex_hull(pts);
64     int m = hull.size();
65     double ans = 0;
66     // 旋轉卡殼
67     for (int i = 0; i < m; ++i) {
68         int j = (i + 1) % m, k = (j + 1) % m;
69         do {
70             while (true) {

```

```

57         int k2 = (k + 1) % m;
58         double r1 = circumradius(hull[i], hull[j]);
59         double r2 = circumradius(hull[i], hull[j],
60         ↪ hull[k2]);
61         if (r2 > r1) k = k2;
62         else break;
63     }
64     ans = max(ans, circumradius(hull[i], hull[j],
65     ↪ hull[k]));
66     j = (j + 1) % m;
67     } while (j != i);
68 }
cout << fixed << setprecision(6) << ans << endl;
}

```

極角排序.cpp

```

1 struct Point {
2     long long x, y;
3 };
4
5 // 以 base 為原點極角排序：從+x軸(角度0)逆時針到2；
6 ↪ 同角度距離近的在前
7 // 必須先分上下半平面再比cross，
8 ↪ 否則超過180度時comparator無效(UB)
9 // base 本身若在點集中會被排在最前(視為角度0距離0)
10 void polarSort(vector<Point>& points, Point base) {
11     auto half = [&](const Point& p) { // 0:上半平面(含+x軸),
12     ↪ 1:下半平面(含-x軸)
13         long long dx = p.x - base.x, dy = p.y - base.y;
14         return (dy < 0 || (dy == 0 && dx < 0)) ? 1 : 0;
15     };
16     sort(points.begin(), points.end(), [&](const Point& a,
17     ↪ const Point& b) {
18         if (half(a) != half(b)) return half(a) < half(b);
19         long long ax = a.x - base.x, ay = a.y - base.y;
20         long long bx = b.x - base.x, by = b.y - base.y;
21         long long cr = ax * by - ay * bx;
22         if (cr != 0) return cr > 0;
23         return ax * ax + ay * ay < bx * bx + by * by; //
24         ↪ 座標<=1e9不溢位
25     });
26 }

```

5 Data Structure

Array Version of Linked List.cpp

```

1 // 陣列版雙向串列。用前必須先 init 串好(0 與 n+1 是 sentinel)
2 ↪ 資料放 1..n,
3 // 否則 cut/insert 解參考 NULL 直接 segfault. cut(p): 把 p
4 ↪ 從串列拔出並回傳;
5 // insert(p,q): 把 q 插到 p 右邊. sentinel 本身不可被 cut.
6 struct node{
7     struct node *nxt, *prv;
8     long long v;
9 }pos[100002];
10
11 void init(int n){ // 0 -> 1 -> ... -> n -> n+1
12     for(int i=0;i<=n+1;i++){
13         pos[i].nxt = (i<=n ? &pos[i+1] : NULL);
14         pos[i].prv = (i>=1 ? &pos[i-1] : NULL);
15     }
16 }
17
18 inline node* cut(node *p) // cut the node
19 {
20     p->nxt->prv = p->prv;
21     p->prv->nxt = p->nxt;
22     p->nxt = p->prv = NULL;
23     return p;
24 }

```

```

}
inline node* insert(node *p, node *q) // insert node q to
↪ right side of node p
{
    q->nxt = p->nxt;
    p->nxt->prv = q;
    q->prv = p;
    p->nxt = q;
    return q;
}

```

Binary Index Tree.cpp

```

1 //BIT is always 1 index
2 ll n,t[mxN],a[mxN];
3 void built(){for(i,1,mxN){ t[i]+=a[i]; if(i+(i&-i)<mxN)
4 ↪ t[i+(i&-i)]+=t[i];}} //O(n)
5 void pad(ll p,ll val){for(;p<mxN;t[p]+=val,p+=p&-p);}
6 ll query(ll p){ll ans=0; for(;p;ans+=t[p],p-=p&-p); return
7 ↪ ans;}

```

Disjoint Set Union.cpp

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 class DisjointSets {
5     private:
6         vector<int> parents;
7         vector<int> sizes;
8
9     public:
10         DisjointSets(int size) : parents(size),
11         ↪ sizes(size, 1) {
12             for (int i = 0; i < size; i++) {
13                 parents[i] = i; }
14
15         /** @return the "representative" node in x's
16         ↪ component */
17         int find(int x) {
18             return parents[x] == x ? x : (parents[x] =
19             ↪ find(parents[x]));
20
21         }
22
23         /** @return whether the merge changed connectivity
24         ↪ */
25         bool unite(int x, int y) {
26             int x_root = find(x);
27             int y_root = find(y);
28             if (x_root == y_root) { return false; }
29
30             if (sizes[x_root] < sizes[y_root]) {
31                 swap(x_root, y_root); }
32             sizes[x_root] += sizes[y_root];
33             parents[y_root] = x_root;
34             return true;
35
36         }
37
38         /** @return whether x and y are in the same
39         ↪ connected component */
40         bool connected(int x, int y) { return find(x) ==
41         ↪ find(y); }
42 };

```

Segment Tree.cpp

```

1 //range add range query // sum query
2 #define forr(a,b,c) for(ll a=b;a<=c;++a)
3 #define forn(a,b,c) for(ll a=b;a<c;++a)

```

```

4  constexpr ll mxN=2e5+1;
5  #pragma GCC target("lzcnt")
6  ll n,q,t[mxN<<1],lz[mxN]{};
7  void built(){forr(i,n-1,1)t[i]=t[i<<1]+t[i<<1|1];}
8  void apply(ll p,ll k,ll val){t[p]+=val*k; if(p<n)
   ↳ lz[p]+=val;}
9  void push(ll p){for(ll h=63-__builtin_clzll(p);h-->0)
   ↳ if(lz[p>>h]){apply((p>>h)<<1,1<<(h-1),lz[p>>h]),apply(
   ↳ (p>>h)<<1|1,1<<(h-1),lz[p>>h]),lz[p>>h]=0;}}
10 void built(ll p){for(;;p>>=1;)if(!lz[p])
   ↳ t[p]=t[p<<1]+t[p<<1|1];}
11 void rad(ll l,ll r,ll val){
12     ll lo=l+n-1,ro=r+n-1,k=1; push(lo),push(ro);
13     for(;l<=r;l>>=1,r>>=1,k<<=1){if(l&1) apply(l++,k,val);
   ↳ if(r&1~1) apply(r--,k,val);}
14     built(lo),built(ro);
15 }
16 ll query(ll l,ll r){ ll ans=0; push(l+=n-1),push(r+=n-1);
   ↳ for(;l<=r;l>>=1,r>>=1){if(l&1) ans+=t[l++]; if(r&1~1)
   ↳ ans+=t[r--];} return ans;} //range query
17 ll query(ll p){push(p+=n-1); return t[p];} // point query

```

persistence-seg-tree.cpp

```

1  // 節點數: build用 $2n-1$ , 每次單點改約 $\log(n)+1$ 個;  $mxT$ 要  $\geq$ 
   ↳  $2n+q*20$  ( $n=q=2e5$ 時約 $4.2e6$ )
2  // 記憶體約  $7e6*(8+4+4)$  106MB; 多測時  $t/lc/rc$  殘留值會錯,
   ↳ 需清到  $sz$  為止
3  constexpr ll mxN=2e5+1;
4  constexpr ll mxT=7e6+1;
5  ll n,q,t[mxT],a[mxN],rt[mxN<<1],rsc,sz; int
   ↳ lc[mxT],rc[mxT];
6  void built(ll i,ll sl,ll sr){
7     if(sl==sr){t[i]=a[sl]; return;}
8     ll mid; mid=(sl+sr)>>1;
9     lc[i]=++sz,rc[i]=++sz;
10    built(lc[i],sl,mid);
11    built(rc[i],mid+1,sr);
12    t[i]=t[lc[i]]+t[rc[i]];
13 }
14 ll pst(ll pos,ll val,ll i,ll sl,ll sr){
15     ll x; x=++sz;
16     if(sl==sr){t[x]=val; return x;}
17     ll mid; mid=(sl+sr)>>1;
18     if(pos<=mid)
19         ↳ lc[x]=pst(pos,val,lc[i],sl,mid),rc[x]=rc[i];
20     else lc[x]=lc[i],rc[x]=pst(pos,val,rc[i],mid+1,sr);
21     t[x]=t[lc[x]]+t[rc[x]];
22     return x;
23 }
24 ll query(ll l,ll r,ll i,ll sl,ll sr){
25     if(l>sr || r<sl) return 0;
26     if(l<=sl && sr<=r) return t[i];
27     ll mid; mid=(sl+sr)>>1;
28     return query(l,r,lc[i],sl,mid)+query(l,r,rc[i],mid+1,sr);
29 }
30 void solve(istream &cin){
31     cin>>n>>q;
32     sz=rsc=0;
33     forn(i,1,n+1) cin>>a[i];
34     rt[++rsc]=++sz;
35     built(rt[1],1,n);
36     while(q--){
37         ll op; cin>>op;
38         if(op==3){
39             ll k; cin>>k;
40             rt[++rsc]=++sz;
41             lc[rt[rsc]]=lc[rt[k]],rc[rt[rsc]]=rc[rt[k]];
42             ↳ rc[rt[k]],t[rt[rsc]]=t[rt[k]];
43         }
44         else if(op==1){
45             ll k,a,x; cin>>k>>a>>x;
46             rt[k]=pst(a,x,rt[k],1,n);
47         }
48         else{

```

```

47     ll k,a,b; cin>>k>>a>>b;
48     cout << query(a,b,rt[k],1,n) <<
   ↳ '\n';
49 }
50 }
51 }
52 }

```

splay-tree.cpp

```

1  constexpr ll mxN=2e5+3;
2  ll n,m,rt,t[mxN][2]{},cnt[mxN],val[mxN],f[mxN],lz[mxN]{};v
   ↳ als[mxN];
3  string s;
4  ll dir(ll x){return t[f[x]][1]==x;}
5  void push_down(ll x){
6     if(lz[x]){
7         swap(t[x][0],t[x][1]);
8         if(t[x][0]) lz[t[x][0]]^=1;
9         if(t[x][1]) lz[t[x][1]]^=1;
10        lz[x]=0;
11    }
12 }
13 void push(ll x){cnt[x]=cnt[t[x][0]]+1+cnt[t[x][1]],vals[x]
   ↳ =vals[t[x][0]]+val[x]+vals[t[x][1]];}
14 void rotate(ll x){
15     ll y=f[x],z=f[y],d=dir(x);
16     t[y][d]=t[x][d^1],t[x][d^1]=y;
17     if(t[y][d]) f[t[y][d]]=y;
18     if(z) t[z][dir(y)]=x;
19     f[y]=x,f[x]=z,push(y),push(x);
20 }
21 void splay(ll &z,ll x){
22     ll w=f[z];
23     for(ll y;(y=f[x])!=w;rotate(x)){
24         if(f[y]!=w) rotate(dir(x)==dir(y)?y:x);
25     }
26     z=x;
27 }
28 void build(){
29     rt=0;
30     forn(i,1,n+3){
31         t[i][0]=rt;
32         if(rt) f[rt]=i;
33         rt=i;
34     }
35     splay(rt,1);
36 }
37 void loc(ll &z,ll k){
38     ll x=z;
39     for(push_down(x);cnt[t[x][0]]+1!=k;push_down(x)){
40         if(k>cnt[t[x][0]])
41             ↳ k-=cnt[t[x][0]]+1,x=t[x][1];
42         else x=t[x][0];
43     }
44     splay(z,x);
45 }
46 void opt(ll l,ll r){
47     loc(rt,l);
48     loc(t[rt][1],r-1+2);
49     ll x=t[t[rt][1]][0];
50     lz[x]^=1;
51 }
52 ll query(ll l,ll r){
53     loc(rt,l),loc(t[rt][1],r-1+2);
54     ll x=t[t[rt][1]][0],y=vals[x];
55     return y;
56 }
57 void solve(istream &cin){
58     cin >> n >> m;
59     forn(i,2,n+2) cin >> val[i];
60     build();
61     forn(i,0,m){
62         ll op,l,r; cin >> op >> l >> r;
63         if(op==1) opt(l,r);
64         else cout << query(l,r) << '\n';

```



```

64     }
65 }

```

6 Not that important

EPS.cpp

```

1  #include <cmath>
2  const double EPS = 1e-9;
3
4  bool eq(double a, double b) {
5      return fabs(a - b) < EPS;
6  }
7
8  bool lt(double a, double b) {
9      return a < b - EPS;
10 }
11
12 bool le(double a, double b) {
13     return a < b + EPS;
14 }

```

Gauce Elimination.cpp

```

1  // a[N][N] 是增广矩阵
2  int gauss()
3  {
4      int c, r;
5      for (c = 0, r = 0; c < n; c++)
6      {
7          int t = r;
8          for (int i = r; i < n; i++) //
9              ↪ 找到绝对值最大的行
10             if (fabs(a[i][c]) > fabs(a[t][c]))
11                 t = i;
12
13             if (fabs(a[t][c]) < eps) continue;
14
15             for (int i = c; i <= n; i++) swap(a[t][i],
16             ↪ a[r][i]); // 将绝对值最大的行换到最顶端
17             for (int i = n; i >= c; i--) a[r][i] /= a[r][c];
18             ↪ // 将当前行的首位变成1
19             for (int i = r + 1; i < n; i++) //
20                 ↪ 用当前行将下面所有的列消成0
21                 if (fabs(a[i][c]) > eps)
22                     for (int j = n; j >= c; j--)
23                         a[i][j] -= a[r][j] * a[i][c];
24
25             r++;
26     }
27
28     if (r < n)
29     {
30         for (int i = r; i < n; i++)
31             if (fabs(a[i][n]) > eps)
32                 return 2; // 无解
33         return 1; // 有无穷多组解
34     }
35
36     for (int i = n - 1; i >= 0; i--)
37         for (int j = i + 1; j < n; j++)
38             a[i][n] -= a[i][j] * a[j][n];
39
40     return 0; // 有唯一解
41 }

```

Joseph.cpp

```

1  // 回傳 0-indexed 的倖存者位置
2  ll josephus(ll n, ll k) { // O(n)

```

```

3      ll res = 0;
4      for (ll i = 1; i <= n; ++i) res = (res + k) % i;
5      return res;
6  }
7  ll josephus2(ll n, ll k) { // O(k log n), k<=n 時用; k>n
8      ↪ 落回迴圈版避免深遞迴爆棧
9      if (n == 1) return 0;
10     if (k == 1) return n - 1;
11     if (k > n) {
12         ll res = 0;
13         for (ll i = 2; i <= n; ++i) res = (res + k) % i;
14         return res;
15     }
16     ll res = josephus2(n - n / k, k);
17     res -= n % k;
18     if (res < 0)
19         res += n; // mod n
20     else
21         res += res / (k - 1); // 还原位置
22     return res;
23 }

```

7 New

Convex_hull_trick.cpp

```

1  // 特定題解法: 斜率=(y2-y1)/m 是整數除法(負數向零截斷),
2  ↪ 查詢x需單調遞增
3  // check() 乘法沒用__int128, c差*m差 ~1e18 以上會溢位,
4  ↪ 改編他題時注意
5  class line{
6      public:
7          ll m,c;
8          line(){}
9          line(ll m,ll c):m(m),c(c){}
10         friend bool operator<(const line &a,const
11         ↪ line &b){
12             return a.m<b.m || (a.m==b.m &&
13             ↪ a.c>b.c);
14         }
15         ll cal(ll x){return m*x+c;}
16     };
17     constexpr ll mxN=1e5+1;
18     ll n,m,id=0,st[mxN];
19     vc<line> ln;
20     bool check(ll i,ll j,ll k){
21         return (ln[i].c-ln[k].c)*(ln[j].m-ln[i].m)<(ln[i].c-
22         ↪ ln[j].c)*(ln[k].m-ln[i].m);
23     }
24     void solve(istream &cin){
25         cin >> n >> m;
26         forn(i,0,n){
27             ll y1,y2; cin >> y1 >> y2;
28             ln.emplace(y2-y1/m,y1);
29         }
30         sort(all(ln)),ln.resize(unique(all(ln),[](auto
31         ↪ a,auto b){return a.m==b.m;})-ln.bg());
32         forn(i,0,ln.size()){
33             while(id>1 && check(st[id-1],st[id],i))
34                 ↪ id--;
35             st[++id]=i;
36         }
37         ll ptr=1;
38         forn(i,0,m+1){
39             while(ptr+1<id && ln[st[ptr+1]].cal(i)>ln[
40             ↪ st[ptr]].cal(i))
41                 ↪ ptr++;
42             cout << ln[st[ptr]].cal(i) << " \n"[i==m];
43         }
44     }
45 }

```


Ray_casting.cpp

```

1  class Point{
2      public:
3          ll x,y;
4          Point(){ }
5          Point(ll x,ll y):x(x),y(y){ }
6          friend Point operator-(const Point
            ↪ &a,const Point &b){return
            ↪ {a.x-b.x,a.y-b.y};}
7          friend ll operator*(const Point &a,const
            ↪ Point &b){return a.x*b.x+a.y*b.y;}
8          friend ll operator^(const Point &a,const
            ↪ Point &b){return a.x*b.y-a.y*b.x;}
9      };
10     ll n,m;
11     vc<Point> p;
12     bool on(Point &a,Point &b,Point &c){
13         if((b-a)^(c-a)) return false;
14         return 0<=((b-a)*(c-a)) &&
            ↪ ((b-a)*(c-a))<=((b-a)*(b-a));
15     }
16     void query(Point &x){
17         ll ok=0;
18         forn(i,0,n){
19             if(on(p[i],p[(i+1)%n],x)){ cout <<
            ↪ "BOUNDARY" << '\n';return;}
20             Point &a=p[i],&b=p[(i+1)%n];
21             if(a.y>x.y != b.y>x.y){
22                 ll sg=(x-a)^(b-a);
23                 if(b.y>a.y==(sg<0)) ok+=1;
24             }
25         }
26         cout << (ok&1?"INSIDE":"OUTSIDE") << '\n';
27     }

```

block_cut_tree.cpp

```

1  constexpr ll mxN=1e5+1;
2  constexpr ll mxB=20;
3  ll n,m,q,st[mxN],id=0,mark[mxN]{},dfn[mxN]{},low[mxN],tim=
            ↪ 0,nid[mxN],pa[mxN<<1][mxB],d[mxN<<1];
4  vc<ll> adj[mxN];
5  vc<vc<ll>> cmp,t;
6  void tarjan(ll u=1,ll p=0){
7      dfn[u]=low[u]=++tim,st[++id]=u;
8      for(auto&v:adj[u]){
9          if(v==p) continue;
10         if(dfn[v]) low[u]=min(low[u],dfn[v]);
11         else{
12             tarjan(v,u);
13             low[u]=min(low[u],low[v]);
14             if(low[v]>dfn[u]){
15                 mark[u]=(dfn[u]>1 ||
                    ↪ dfn[v]>2);
16                 cmp.emp(vc<ll>{u});
17                 while(cmp.back().back()!=v){
18                     ↪ )
19                     ↪ cmp.back().emp(st[id--]);
20                 ↪ ];
21             }
22         }
23     }
24     void dfs(ll u=1,ll p=0,ll lv=1){
25         d[u]=lv,pa[u][0]=p;
26         for(auto&v:t[u]){
27             if(v!=p) dfs(v,u,lv+1);
28         }
29     }
30     void built(){
31         ll x=0;
32         t.emp(vc<ll>{});
33         forn(i,1,n+1) if(mark[i])
            ↪ nid[i]=++x,t.emp(vc<ll>{});
34         for(auto&a:cmp){

```

```

35         ll y=++x;
36         t.emp(vc<ll>{});
37         for(auto&u:a){
38             if(mark[u]) t[nid[u]].emp(y),t[y].
            ↪ emp(nid[u]);
39             else nid[u]=y;
40         }
41         dfs();
42         forn(j,1,mxB) forn(i,1,t.size())
            ↪ pa[i][j]=pa[pa[i][j-1]][j-1];
43     }
44     ll lca(ll a,ll b){
45         if(d[a]<d[b]) swap(a,b);
46         forr(i,mxB-1,0) if(d[a]-(1<<i)>=d[b]) a=pa[a][i];
47         if(a==b) return a;
48         forr(i,mxB-1,0) if(pa[a][i]!=pa[b][i])
            ↪ a=pa[a][i],b=pa[b][i];
49         return pa[a][0];
50     }
51     ll query(ll a,ll b,ll c){
52         ll lca1=lca(a,b),lca2=lca(a,c),lca3=lca(b,c);
53         return (lca1==c || (lca2==c && lca3==lca1) ||
            ↪ (lca3==c && lca2==lca1));
54     }
55     void solve(){
56         cin >> n >> m >> q;
57         forn(i,0,m){
58             ll u,v; cin >> u >> v;
59             adj[u].emp(v),adj[v].emp(u);
60         }
61         tarjan();
62         built();
63         while(q--){
64             ll a,b,c; cin >> a >> b >> c;
65             cout << (mark[c] &&
            ↪ query(nid[a],nid[b],nid[c])) || a==c
            ↪ || b==c?"NO":"YES" << '\n';
66         }

```

cross_dot_product.cpp

```

1  //A=(x1,y1),B=(x2,y2)
2  //A cross B = ABsin x = x1*y2-y1*x2
3  //A dot B =AB cos x = x1*x2+y1*y2

```

dec_string_to_bitset.cpp

```

1  // 10進位string(只含0-9, 長度<=500) 轉 bitset 二進位,
            ↪ 支援先減一個小常數
2  // decSub(s,x): 回傳 s-x (x>=0, 需保證 s>=x), 0(len)
3  // decToBits(s): 10進string -> bitset, b[0]為LSB, 0(len^2)
4  // bitsToDec(b): bitset -> 10進 string, 驗證/輸出用
5  // 500位 10進 1661 bits, mxB=2048 夠用
6  constexpr ll mxB=2048;
7  string decSub(string s,ll x){
8       for(ll i=s.size()-1;i>=0&&x;i--){
9           ll d=(s[i]-'0')-x%10; x/=10;
10          if(d<0){d+=10; x++;}
11          s[i]=d+'0';
12      }
13      ll p=0; while(p+1<(ll)s.size()&&s[p]=='0') p++;
14      return s.substr(p);
15  }
16  bitset<mxB> decToBits(string s){
17      bitset<mxB> b;
18      for(ll i=0;s!="0";i++){
19          b[i]=(s.back()-'0')&1;
20          ll r=0;
21          for(char &c:s){ll d=r*10+(c-'0');
            ↪ c='0'+d/2; r=d&1;}
22          ll p=0; while(p+1<(ll)s.size()&&s[p]=='0')
            ↪ p++;

```

```

23         s=s.substr(p);
24     }
25     return b;
26 }
27 string bitsToDec(bitset<mxB> b){
28     string s="0";
29     ll hi=mxB-1; while(hi>0&&!b[hi]) hi--;
30     forr(i,hi,0){
31         ll c=b[i];
32         forr(j,(ll)s.size()-1,0){ll
            ↪ d=(s[j]-'0')*2+c; s[j]='0'+d%10;
            ↪ c=d/10;}}
33         if(c) s=char('0'+c)+s;
34     }
35     return s;
36 }
37 void solve(istream &cin){
38     string s; ll k;
39     cin>>s>>k; // 10 進 string 與 要減的常數
40     bitset<mxB> b=decToBits(decSub(s,k));
41     ll hi=mxB-1; while(hi>0&&!b[hi]) hi--;
42     forr(i,hi,0) cout<<b[i]; // MSB->LSB 印出 2 進位
43     cout<<'\n';
44 }

```

dominator_tree.cpp

```

1 constexpr ll mxN=1e5+1;
2 ll n,m,tim=0,id[mxN],sd[mxN],pa[mxN],f[mxN],lb[mxN],et[mxN]
    ↪ [] {},ret[mxN];
3 vc<ll> adj[mxN],radj[mxN],t[mxN],st;
4 void dfs(ll u=1){
5     et[u]++tim,ret[tim]=u;
6     id[tim]=sd[tim]=lb[tim]=pa[tim]=tim;
7     for(auto&v:adj[u]){
8         if(!et[v]) dfs(v),f[et[v]]=et[u];
9         radj[et[v]].emp(et[u]);
10    }
11 }
12 ll find(ll u,ll x=0){
13     if(u==pa[u]) return x?-1:u;
14     ll v=find(pa[u],x+1);
15     if(v<0) return u;
16     if(sd[lb[u]]>sd[lb[pa[u]]]) lb[u]=lb[pa[u]];
17     pa[u]=v;
18     return x?v:lb[u];
19 }
20 void built(){
21     vc<ll> arr[n+1];
22     forr(i,n,1){
23         for(auto&v:radj[i])
            ↪ sd[i]=min(sd[i],sd[find(v)]);
24         if(i>1) arr[sd[i]].emp(i);
25         for(auto&v:arr[i]){
26             ll x=find(v);
27             if(sd[x]==sd[v]) id[v]=sd[v];
28             else id[v]=x;
29         }
30         if(i>1) pa[i]=f[i];
31     }
32     forn(i,2,n+1){
33         if(id[i]!=sd[i]) id[i]=id[id[i]];
34         t[ret[i]].emp(ret[id[i]]),t[ret[id[i]]].emp
            ↪ p(ret[i]);
35     }
36 }
37 void dfs_ans(ll u=1,ll p=0){
38     st.emp(u);
39     if(u==n){
40         sort(all(st));
41         cout << st.size() << '\n';
42         for(auto&v:st) cout << v << ' '; cout <<
            ↪ '\n';
43         exit(0);
44     }
45     for(auto&v:t[u]){
46         if(v==p) continue;

```

```

47         dfs_ans(v,u);
48     }
49     st.pop_back();
50 }
51 void solve(istream &cin){
52     cin>>n>>m;
53     forn(i,1,m+1){
54         ll u,v; cin>>u>>v;
55         adj[u].emp(v);
56     }
57     dfs();
58     built();
59     dfs_ans();
60 }

```

extra.cpp

```

1 //__builtin_popcountll
2 //__builtin_parityll
3 //__builtin_clzll
4 //__builtin_ctzll
5 // mt19937 rng(chrono::steady_clock::now().time_since_epoc
    ↪ h().count());
6 // const ll M = 991831889;
7 // const ll C = uniform_int_distribution<ll>(0.1 * M, 0.9 *
    ↪ M)(rng);

```

fft.cpp

```

1 class FFT{
2     public:
3         ll mod,root,root_inv,maX,M,M_inv;
4         FFT(ll a,ll b,ll c,ll d,ll e,ll
            ↪ f):mod(a),root(b),root_inv(c),maX(d),M
            ↪ (e),M_inv(f){}
5         ll fsp(ll a,ll m){
6             ll ans=1;
7             while(m){
8                 if(m&1) ans=(ans*a)%mod;
9                 a=(a*a)%mod,m>>=1;
10            }
11            return ans;
12        }
13    }
14    void fft(ll sz,vc<ll> &arr,ll inv){
15        for(ll i=1,j=0;i<sz;++i){
16            ll bit=sz>>1;
            ↪ for(;j&bit;bit>>=1)
            ↪ j^=bit; j^=bit;
17            if(i<j)
            ↪ swap(arr[i],arr[j]);
18        }
19        for(ll l=2;l<=sz;l<=<=1){
20            ll wlen=inv?root_inv:root;
21            for(ll i=l;i<maX;i<=<=1)
            ↪ wlen=(wlen*wlen)%mod;
22            for(ll i=0;i<sz;i+=l){
23                ll w=1;
24                for(ll
                    ↪ j=0;j<l/2;++j){
25                    ll x=arr[i
                        ↪ +j],y=
                        ↪ (w*arr
                        ↪ [i+j+l
                        ↪ /2])%m
                        ↪ od;
26                    arr[i+j]=(
                        ↪ x+y<mo
                        ↪ d?x+y:
                        ↪ x+y-mo
                        ↪ d);

```

```

27         arr[i+j+1/4]
        ↪ 2]=(x-
        ↪ y>=0?x5
        ↪ -y:x-y6
        ↪ +mod);7
28     w=(w*wlen)
        ↪ %mod; 8
        }
29     }
30     }
31     }
32     if(inv){
33         ll n_1=fsp(sz,mod-2);
34         forn(i,0,sz){
35             arr[i]=(arr[i]*n_1
        ↪ )%mod;
36         }
37     }
38 }
39 };
40 constexpr ll MM=1002772198720536577;
41 //
42 //g=3
43 //998244353=119*2^23 + 1
44 //g^c=15311432
45 //inv=469870224
46 //M1=1004535809
47 //M1^-1=332747959
48 //
49 //1004535809=479*2^21 + 1
50 //g^c=702606812
51 //inv=700146880
52 //M2=998244353
53 //M2^-1=669690699
54 ll mul(ll a,ll b){
55     __int128 aa=a,bb=b,cc;
56     cc=(aa*bb)%MM;
57     return (ll)cc;
58 }
59 void solve(){
60     FFT fft1(998244353,15311432,469870224,1<<23,100453
        ↪ 5809,332747959);
61     FFT fft2(1004535809,702606812,700146880,1<<21,9982
        ↪ 44353,669690699);
62     //ll x=(fft1.M*fft1.M_inv)%fft1.mod;
63     //ll y=(fft2.M*fft2.M_inv)%fft2.mod;
64     //err(x,y);
65     ll n,m;
66     cin >> n >> m;
67     vc<ll> a(n+1),b(m+1);
68     for(auto &v:a) cin >> v;
69     for(auto &v:b) cin >> v;
70     ll sz=1;
71     for(;sz<a.size()+b.size();sz<=1);
72     a.resize(sz),b.resize(sz);
73     vc<ll> A=a,B=b;
74     fft1.fft(sz,a,0);
75     fft1.fft(sz,b,0);
76     forn(i,0,sz) a[i]=(a[i]*b[i])%fft1.mod;
77     fft1.fft(sz,a,1);
78
79     fft2.fft(sz,A,0);
80     fft2.fft(sz,B,0);
81     forn(i,0,sz) A[i]=(A[i]*B[i])%fft2.mod;
82     fft2.fft(sz,A,1);
83
84     forn(i,0,n+m+1) cout <<
        ↪ (mul(a[i],fft1.M*fft1.M_inv) +
        ↪ mul(A[i],fft2.M*fft2.M_inv))%MM << "
        ↪ \n"[i==n+m];
85 }

```

fwf.cpp

```

1  constexpr ll mxB=21;
2  ll a[1<<mxB]{};
3  void fwf(ll *arr,ll type){

```

```

for(ll k=2;k<=(1<<mxB);k<=1){ //必須<=,
    ↪ 否則最高層蝶形沒做, 值>=2^(mxB-1)會錯
        forn(ll i=0,m=k>>1;i+m<(1<<mxB);i+=k){
            forn(j,0,m){
                ll x,y; x=arr[i+j],y=arr[i
                ↪ +j+m];
                arr[i+j]=x+y,arr[i+j+m]=x-
                ↪ y;
                if(type) arr[i+j]/=2,arr[i
                ↪ +j+m]/=2;
            }
        }
    }
}

void solve(istream &cin){
    ll n,pfx,mx;
    cin>>n;
    mx=pfx=0;
    a[0]=1;
    forn(i,0,n){
        ll x; cin>>x;
        pfx^=x,a[pfx]++,mx=max(mx,a[pfx]);
    }
    fwf(a,0);
    forn(i,0,1<<mxB) a[i]=a[i]*a[i];
    fwf(a,1);
    ll res; res=0;
    // forn(i,0,17) cout << a[i] << " \n"[i==16];
    forn(i,0,1<<mxB) res+=(!i?mx>1:a[i]>0);
    cout << res << '\n';
    forn(i,0,1<<mxB) if(!i?mx>1:a[i]) cout << i << '
    ↪ '; cout << '\n';
}

```

kmp-automaton.cpp

```

void kmp(){
    forn(i,1,m){
        ll j; j=k[i-1];
        while(j>0 && s[j]!=s[i]) j=k[j-1];
        if(s[j]==s[i]) k[i]=j+1;
    }
}

void solve(istream &cin){
    cin>>n>>s;
    m=s.length();
    kmp();
    //forn(i,0,m){err(i,k[i]);}
    forn(i,0,m+1){
        forn(j,0,26){
            if(i>0 && (s[i]-'A')!=j)
                ↪ nex[i][j]=nex[k[i-1]][j];
            else nex[i][j]=i+(s[i]-'A'==j);
        }
    }
    // forn(i,0,m+1){forn(j,0,26){err(i,j,nex[i][j]);}}
    dp[0][0][0]=1;
    forn(i,0,n){
        forn(f,0,2){
            forn(j,0,m+1){
                forn(c,0,26){
                    // err(i,j,c,nex[j
                    ↪ ][c],dp[i][j]);
                    ll nf; nf=f|(nex[j
                    ↪ ][c]==m);
                    dp[i+1][nex[j][c]][
                    ↪ nf]=(dp[i+1][
                    ↪ nex[j][c]][nf]
                    ↪ +dp[i][j][f])%
                    ↪ md11;
                }
            }
        }
    }
}

```

```

32     ll res; res=0;
33     forn(i,0,m+1) res=(res+dp[n][i][1])%md11;
34     cout << res << '\n';
35 }
36

```

li_chao_tree.cpp

```

1  class line {
2      public:
3          ll m,c;
4          line(){m=0,c=-inf-1;}
5          line(ll m,ll c):m(m),c(c){}
6          ll cal(ll x){return m*x+c;}
7  };
8  constexpr ll mxN=1e5+5;
9  ll n;
10 line t[mxN<<2];
11 void insert(line x,ll l,ll r,ll i,ll sl,ll sr){
12     if(sr<l || r<sl || l>r) return;
13     ll mid=(sl+sr)>>1;
14     if(l<=sl && sr<=r){ //if extended otherwise remove
15         ↪ this condition
16         bool lf=t[i].cal(sl-1)<x.cal(sl-1),mf=t[i].cal(sl-1)<x.cal(mid-1),rf=t[i].cal(sr-1)<x.cal(sr-1);
17         if(mf) swap(x,t[i]);
18         if(sl==sr) return;
19         if(lf^mf) insert(x,l,r,i<<1,sl,mid);
20         if(rf^mf) insert(x,l,r,i<<1|1,mid+1,sr);
21         return;
22     }
23     insert(x,l,r,i<<1,sl,mid);
24     insert(x,l,r,i<<1|1,mid+1,sr);
25 }
26 ll query(ll x,ll i,ll sl,ll sr){
27     ll mid=(sl+sr)>>1;
28     if(sl==sr) return t[i].cal(x-1);
29     if(x<=mid) return
30     ↪ max(t[i].cal(x-1),query(x,i<<1,sl,mid));
31     else return
32     ↪ max(t[i].cal(x-1),query(x,i<<1|1,mid+1,sr));
33 }
34 void solve(){
35     cin >> n;
36     forn(i,0,n){
37         ll op;
38         cin >> op;
39         if(op==1){
40             ll a,b,l,r;
41             cin >> a >> b >> l >> r;
42             line x{a,b};
43             insert(x,l+1,r+1,1,1,mxN-4);
44         }
45         else{
46             ll x;
47             cin >> x;
48             x=query(x+1,1,1,mxN-4);
49             if(x>-inf-1) cout << x << '\n';
50             else cout << "NO" << '\n';
51         }
52     }
53 }

```

manacher.cpp

```

1  ll dp[mxN][2],n,res[mxN];
2  void solve(istream &cin){
3      cin >> s,n=s.length();
4      ll l1,l2,r1=-1,r2=-1;
5      forn(i,0,n) res[i]=1;
6      forn(i,0,n){
7          if(r1>i) dp[i][1]=min(r1-i,dp[l1+r1-i][1]);

```

```

        if(r2>i)
        ↪ dp[i][0]=min(r2-i,dp[l2+r2-i][0]);
        while(i+dp[i][1]<n && i-dp[i][1]>=0 &&
        ↪ s[i+dp[i][1]]==s[i-dp[i][1]])
        ↪ res[i+dp[i][1]]=max(res[i+dp[i][1]],2*
        ↪ dp[i][1]+1),dp[i][1]++;
        while(i+dp[i][0]+1<n && i-dp[i][0]>=0 &&
        ↪ s[i+dp[i][0]+1]==s[i-dp[i][0]])
        ↪ res[i+dp[i][0]+1]=max(res[i+dp[i][0]+1],
        ↪ 2*(dp[i][0]+1)),dp[i][0]++;
        if(i+dp[i][1]>r1)
        ↪ l1=i-dp[i][1],r1=i+dp[i][1];
        if(i+dp[i][0]>r2)
        ↪ l2=i-dp[i][0]+1,r2=i+dp[i][0];
    }
    forn(i,0,n) cout << res[i] << " \n"[i==n-1];
}

```

shoelace.cpp

```

//consecutive point p[i] (0<=i<n)
//注意：輸出是「2倍面積」（鞋帶公式沒除2）； 2A =
↪ sum(p[i]^p[i+1])
//座標 |<=1e9 時 2A<=8e18 勉強不溢位，更大要 __int128
void solve(istream &cin){
    ll n,res=0;
    cin>>n;
    forn(i,1,n+1){
        cin>>a[i];
        if(i>1) res+=a[i-1]^a[i]; //cross product
        if(i==n) res+=a[n]^a[1];
    }
    cout << abs(res) << '\n';
}

```

suffix_automaton.cpp

```

1  constexpr ll mxN=1e5+1;
2  class state{
3      public:
4          ll len,link,next[26];
5  };
6  state t{};
7  ll n,sz;
8  state t[mxN<<1];
9  void sam(const string &s){
10     memset(t,0,sizeof(t));
11     t[0].link=-1;
12     ll last,cur; last=0;
13     for(auto&v:s){
14         cur=++sz;
15         t[cur].len=t[last].len+1;
16         ll p; p=last;
17         while(p!=-1 && !t[p].next[v-'a'])
18             ↪ t[p].next[v-'a']=cur,p=t[p].link;
19         if(p==-1) t[cur].link=0;
20         else{
21             ll q; q=t[p].next[v-'a'];
22             if(t[p].len+1==t[q].len) t[cur].link=q;
23             else{
24                 ll clone; clone=++sz;
25                 t[clone].len=t[p].len+1,t[clone].link=t[q].
26                 ↪ .link;
27                 memcpy(t[clone].next,t[q].next,sizeof(t[q].
28                 ↪ .next));
29                 while(p!=-1 && t[p].next[v-'a']==q)
30                     ↪ t[p].next[v-'a']=clone,p=t[p].link;
31                 t[q].link=t[cur].link=clone;
32             }
33         }
34         last=cur;
35     }
36 }

```

```

33 }
34 void solve(istream &cin){
35     string s;
36     cin>>s;
37     n=s.length(),sz=0;
38     sam(s);
39     vc<ll> res(n+2,0);
40     forn(i,1,sz+1)
41         ↪ res[t[t[i].link].len+1]++,res[t[i].len+1]--;
42     partial_sum(all(res),res.bg());
43     forn(i,1,n+1) cout << res[i] << " \n"[i==n];

```

suffixarray_and_lcp.cpp

```

1  constexpr ll mxN=1e5+5;
2  ll sa[mxN],nsa[mxN],c[mxN],cnt[mxN] {},lcp[mxN];
3  void sfa(string s){
4      s+="$";
5      ll n=s.length();
6      memset(cnt,0,sizeof(cnt)); //cnt有殘留,
7      ↪ 不清第二次呼叫必錯
8      forn(i,0,n) cnt[s[i]]++; forn(i,1,256)
9      ↪ cnt[i]+=cnt[i-1];
10     forr(i,n-1,0) sa[--cnt[s[i]]]=i; c[sa[0]]=0;
11     ↪ forn(i,1,n)
12     ↪ c[sa[i]]=c[sa[i-1]]+(s[sa[i]]!=s[sa[i-1]]);
13     for(ll h=0;(1<h)<n;h++){
14         forn(i,0,n) nsa[i]=(c[sa[i]]-(1<h))%n+n)%n;
15         fill(cnt,cnt+c[sa[n-1]]+1,0);
16         forn(i,0,n) cnt[c[nsa[i]]]++;
17         ↪ forn(i,1,c[sa[n-1]]+1)
18         ↪ cnt[i]+=cnt[i-1];
19         forr(i,n-1,0) sa[--cnt[c[nsa[i]]]]=nsa[i];
20         nc[sa[0]]=0;
21         forn(i,1,n){
22             pll pre={c[sa[i-1]],c[(sa[i-1]+(1<h))%n]};
23             ↪ cur={c[sa[i]],c[(sa[i]+(1<h))%n]};
24             nc[sa[i]]=nc[sa[i-1]]+(cur!=pre);
25         }
26         swap(c,nc);
27     }
28 }
29 void Lcp(const string &s){
30     ll n=s.length(),k=0;
31     forn(i,0,n){
32         if(c[i]>n-1){k=0; continue;}
33         ll j=sa[c[i]+1];
34         for(;i+k<n && j+k<n && s[i+k]==s[j+k];k++);
35         lcp[c[i]]=k;
36         if(k) k--;
37     }
38 }

```

xor_basis.cpp

```

1 // 值需 < 2^31; 值域到 1e18 時 mxB 改 60、迴圈上限改 59
2 constexpr ll mxN=2e5+1;
3 constexpr ll mxB=31;
4 ll n,a[mxN],b[mxB]{};
5 void built(){
6     forn(i,1,n+1){
7         ll x; x=a[i];
8         forr(j,30,0){
9             if((x>>j)&1){
10                 if(!b[j]){b[j]=x; break;}
11                 else x^=b[j];
12             }
13         }
14     }
15 }
16 void solve(istream &cin){
17     cin>>n;
18     forn(i,1,n+1) cin>>a[i];
19     built();
20     ll res; res=0;
21     forr(i,30,0){
22         if(res>>i&1^1) res^=b[i];
23     }
24     cout << res << '\n';
25 }

```

zfunction_and_kmp.cpp

```

1 constexpr ll mxN=1e6+1;
2 vc<ll> zf(const string &s){
3     ll n,l,r; n=s.length(),l=r=-1;
4     vc<ll> z(n,0);
5     forn(i,1,n){
6         if(r>i) z[i]=min(z[i-1],r-i);
7         while(i+z[i]<n && s[i+z[i]]==s[z[i]])
8             ↪ z[i]++;
9         if(i+z[i]>r) r=i+z[i],l=i;
10     }
11     return z;
12 }
13 vc<ll> kmp(const string &s){
14     ll n; n=s.length();
15     vc<ll> k(n,0);
16     forn(i,1,n){
17         ll j; j=k[i-1];
18         while(j>0 && s[i]!=s[j]) j=k[j-1];
19         if(s[i]==s[j]) k[i]=j+1;
20     }
21     return k;
22 }

```